Water Quality Analyzer

AAD-OCEMS-600

USER MANUAL

(UM-AAD-OCEMS-600-WQA-2025/01)

AADHAV INTECH
No: 60, 3rd Street, Kasi Estate
West Jafferkhanpet,
Chennai – 600083.
Contact Number: 044 42041270
E Mail: admin@aadhavintech.com

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1. Overview

AAD-OCEMS-600 is a continuous, highly accurate and reliable effluent quality monitoring system which measures Chemical Oxygen Demand (COD), Biochemical Oxygen Demand (BOD), Total Suspended Solids (TSS) and pH using state of the art Dual Wavelength Ultra-Violet (UV) spectrophotometric sensor and electrode sensors. COD-BOD analysis is the measurement of oxygen depletion capacity of a water sample contaminated with effluent/organic waste matter when discharged to a water body. TSS and pH are the measures of undissolved solid particles and the alkalinity/acidity of the water body.

1.1 Method of Operation

The AAD-OCEMS-600 system utilizes the PUVCOD-600 Spectrometric Organic On-Line Analyzer, which operates based on ultraviolet (UV) absorption principles. Specifically, it measures the Spectral Absorption Coefficient at 254 nm (SAC254) a wavelength where many organic compounds strongly absorb UV light.

When water containing soluble organic matter passes through the sensor, the analyser detects the amount of UV light absorbed. This absorption is directly related to the concentration of organic substances present in the water.

Under calibrated conditions, the SAC254 value is mathematically correlated and converted into equivalent COD (Chemical Oxygen Demand) or BOD (Biochemical Oxygen Demand) values. This technique enables real-time, reagent-free monitoring of water quality, making the system highly efficient and environmentally friendly.

By avoiding chemical reagents and using direct optical measurement, the system ensures minimal maintenance, rapid response time, and continuous 24/7 operation suitable for industrial and environmental applications.

1.2 Salient Features

- No reagents, no pollution, more economical and environmental protection
- Small drift, fast response, more accurate measurement
- Standard wavelength guarantees long-term accuracy and stability
- Automatic compensation for turbidity interference
- Auto cleaning by brush to prevent bio-adhesion
- Maintenance-free, long service life and low cost of use
- 7-inch interactive HMI display
- Supports Modbus RTU over RS485 output
- Support Field Calibration methods

2. Technical Specifications

1.	Range	COD	0-500mg/L
		BOD	0-500mg/L
		TSS	0-500mg/L
		pH	0-14
2.	Sensitivity	< 1% FS	
3.	Accuracy	±1 %FS	
4.	Rise Time	≤ 60 s	
5.	Warmup Time	≤ 30 minutes	
6.	Operating Temperature	10 – 40 ° C	
7.	Ambient Humidity	10 – 90 % R.H.	
8.	Installation Method	Immersion Installation	
9.	Output (Analog)	Digital Modbus RTU over RS485	
10.	Sensor Material	High Quality 316L Stainless Steel	
11.	Alarm	1 NO relay contact 24V@1A	
12.	Display	7-inch interactive HMI display	
13.	Power supply	24V@3A	
14.	Production Level	IP68	

3. Installation Procedure

Prerequisites:

- The convenient location for mounting the system must be finalized.
- Power supply adapter (Optional)
- Mounting screws
- User manual
- Screw driver
- Allen Key set

To install the unit at the user premises

1. Power and communication cabling shall be setup up to the panel location.
2. Level the gas detector on solid wall and mark the mounting holes with a pencil.

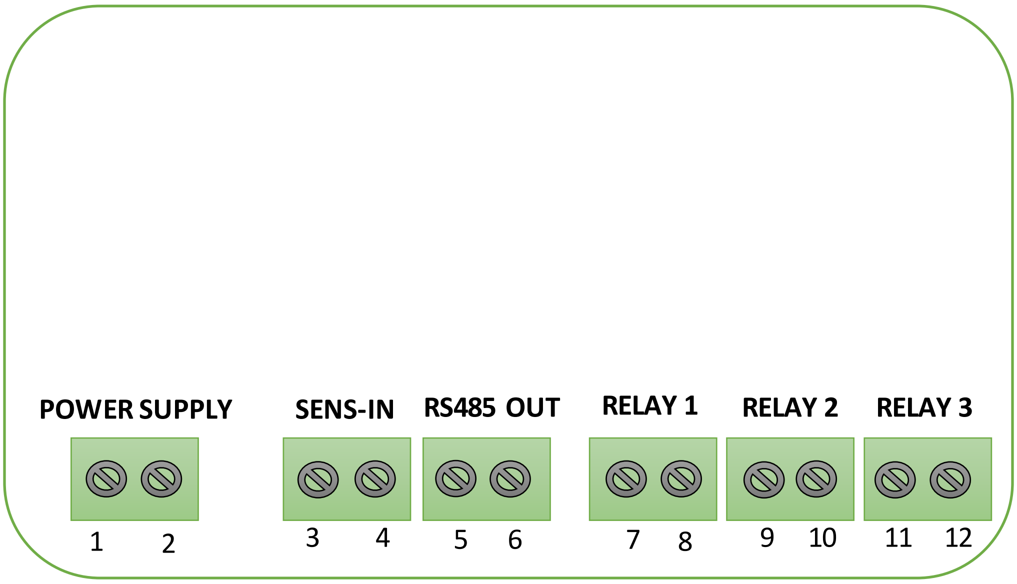
Installation precautions

- The sensor light path should be facing down. Other wise reading will be unreliable

Don'ts

- Don't install the unit in direct sunlight.
- Don't expose the sensor to water concentration higher than its prescribed range.
- Don't attempt to clean the light path.
- Don't open the enclosure with the unit powered ON. Always switch OFF the unit before disassembling.
- Don't switch high voltage or power loads through the relay, use only as potential free contacts.

Connector Details of the unit



CONNECTORS:

Pin No	Description
1	COM
2	24V +
3	RS485 B- (Sensor IN -)
4	RS485 A+ (Sensor IN +)
5	RS485 B- (Output)
6	RS485 A+ (Output)
7	Relay 1 NO
8	Relay 1 COM
9	Relay 2 NO
10	Relay 2 COM
11	Relay 3 NO
12	Relay 3 COM

4. Operation

The AAD-OCEMS-600 system begins with an initial warm-up period of 5 minutes. After this phase, the unit enters scan mode, continuously monitoring the concentration levels of COD, BOD, TSS, and pH in the sample water. The system displays concentration values ranging from 0 to 1000 mg/L. It includes three settable alarm levels, which can be configured from inside the panel. When the measured water quality parameters exceed the configured low or high alarm thresholds, the relay output responds accordingly. The system operates on a supply voltage range of 12V to 24V and features a TFT HMI screen. This user interface allows simple interaction for calibrating the water analyser, setting alarm levels, and configuring communication parameters through a single-touch operation.

Modbus Settings:

Default Settings for AAD-OCEMS-600

Baud Rate	9600
Parity	None
Stop	One
Function code	03
Device Address	1
Data Swapped Float 32 bit	40001 - COD 40003 – BOD 40005 – TSS 40007 – pH 40009 - Temperature

1. HMI (Human Machine Interface) Calibration and Configuration Procedure

Below mentioned image show how to enter the all the mode and calibration procedure

Figure 1 – Loading Screen

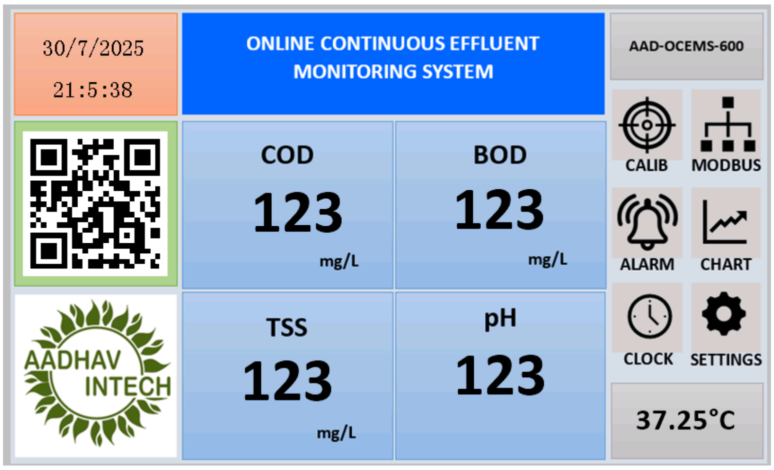


Figure 2 - Home Screen

The Home Screen of the AAD-OCEMS-600 system provides a real-time display of key environmental parameters measured by the effluent monitoring setup, including COD, BOD, TSS (in mg/L), and pH.

At the top, the current date and time are shown to timestamp the readings. A central blue header displays the system title "Online Continuous Effluent Monitoring System" for clear identification. On the left side, a QR code and the Aadhav Intech logo are included for branding and quick access to related information. On the right panel, navigation buttons allow users to access essential functions such as calibration, Modbus testing, alarm settings, data charts, clock configuration, and general settings. The current sample inlet temperature is also displayed in the lower right corner, providing additional context for sensor readings. This user-friendly interface ensures easy monitoring and efficient access to all major features of the system.

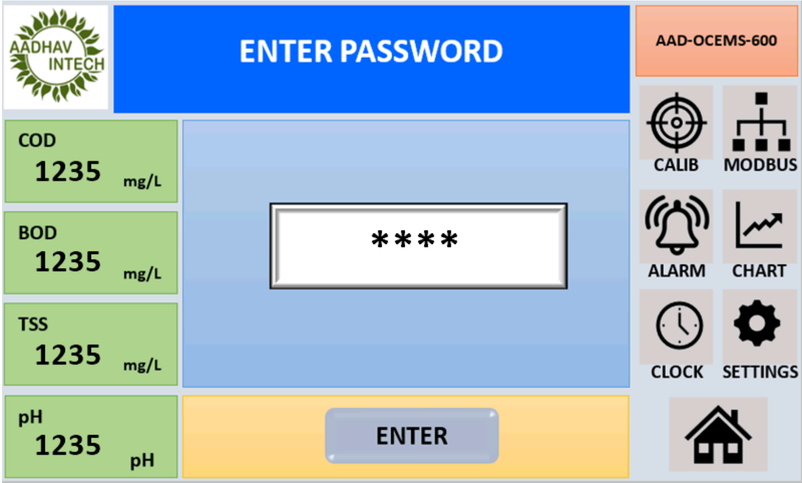


Figure 3 – Password Screen

If you want to go to “CALIBRATION” Screen, press “Calib” Button and enter the correct password and press “ENTER “to navigate the “CALIBRATION” Menu. Same for all the Menu buttons.

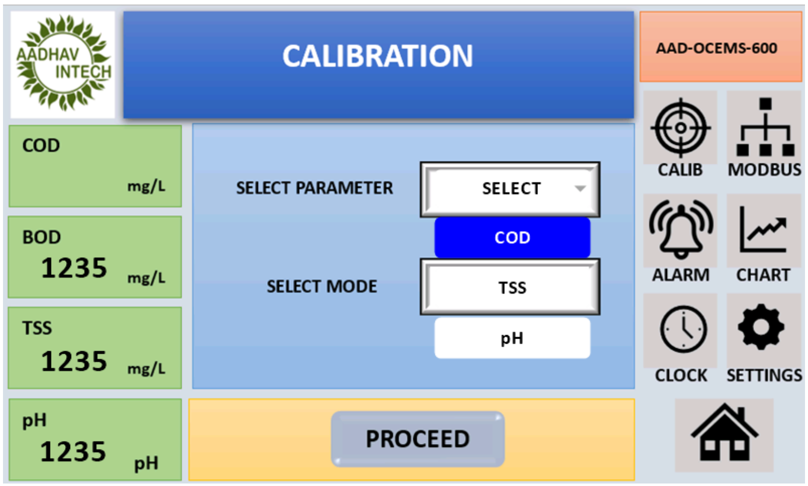


Figure 4 – CALIBRATION SCREEN Select Parameter

On the “CALIBRATION” screen, select the Parameter (“COD, BOD, TSS, Ph”) and choose the “MODE”, Select the Mode (“Zero Solution, 500mg/L and 4.0, 7.0, 9.0)” Ensure the sensor removed from the flow-through Chamber and Slowly immerse the sensor in distilled water to ensure the level of the measuring light path, confirm that the Solution flows into the sensor before pressing the “PROCEED” button to begin calibration.

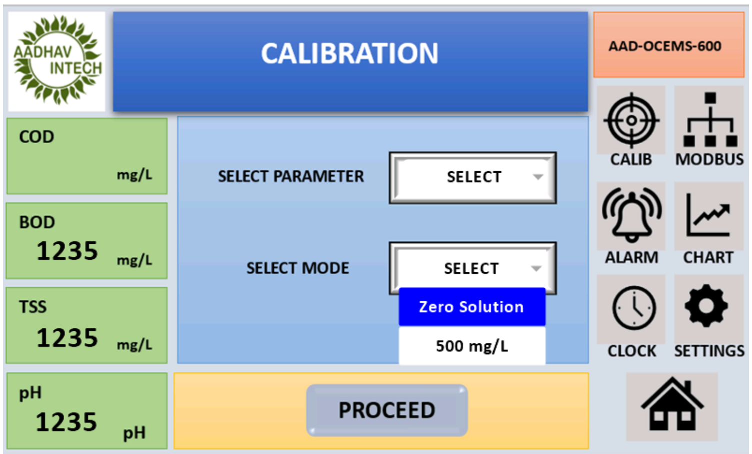


Figure 4 - Select Mode

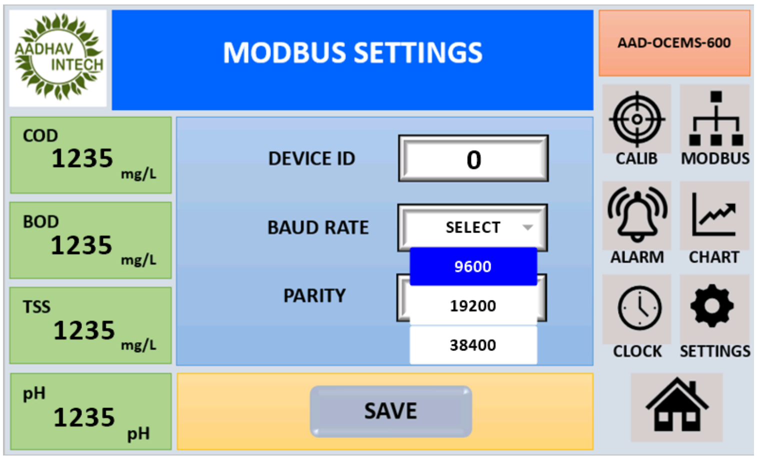


Figure 3 - Modbus Menu

The Modbus Settings screen, The system allows the user to configure communication parameters essential for Modbus RTU-based data exchange with external devices such as PLCs, SCADA systems, or data loggers. The user can set the Device ID (Modbus Slave Address), Baud Rate (communication speed), and Parity (None, Even, or Odd) to match the requirements of the connected Modbus master. These settings must align with the external system for proper communication. After selecting the desired values, pressing the SAVE button will store the configuration and activate the updated settings. Proper configuration of this screen ensures seamless integration and reliable data transmission over RS-485 communication.

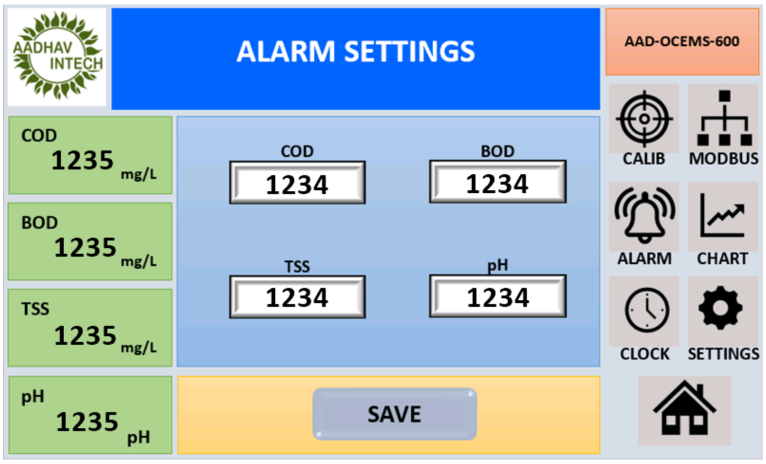


Figure 8 – ALARM SCREEN

On the "ALARM" screen, enter the high-scale alarm values for COD and BOD, TSS, pH then saves the settings to store them in memory. The Alarm Settings screen lets the operator define high- or low-limit thresholds for each effluent parameter—COD, BOD, TSS, and pH—so that the system can trigger alerts when measured values exceed safe ranges.

Current real-time readings are shown along the left side for reference, while editable input fields in the centre allow precise entry of the desired alarm setpoints.

Once the threshold values are entered, pressing the "SAVE" button stores the configuration and activates the monitoring logic. As with other screens, the right-hand panel provides quick navigation icons for Calibration, Modbus, Alarm, Chart, Clock, Settings, and Home, ensuring seamless workflow and easy access to all system functions.

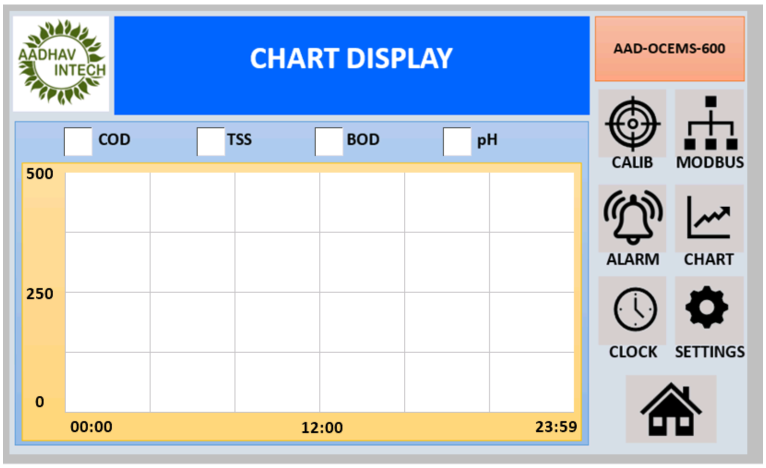


Figure 5 – TREND GRAPH

On the “TREND GRAPH” screen, select the parameter (COD or BOD or TSS or pH) to plot, then press “UPDATE CHART” to display the plotted graph.

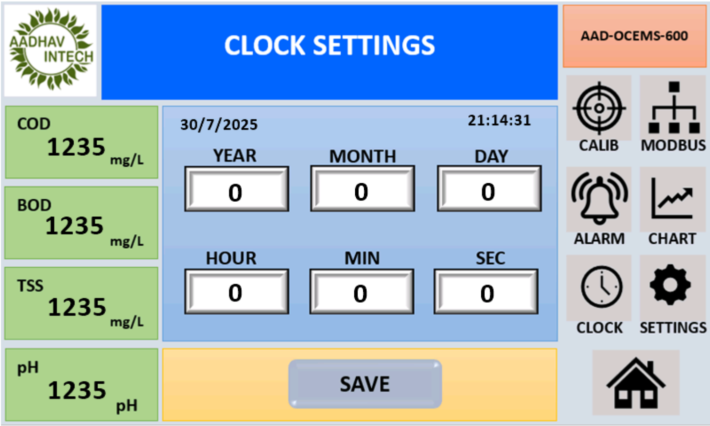


Figure 6 - CLOCK Screen

To adjust the clock, go to the “CLOCK” screen, set the date and time, and press “SET” to save the changes.

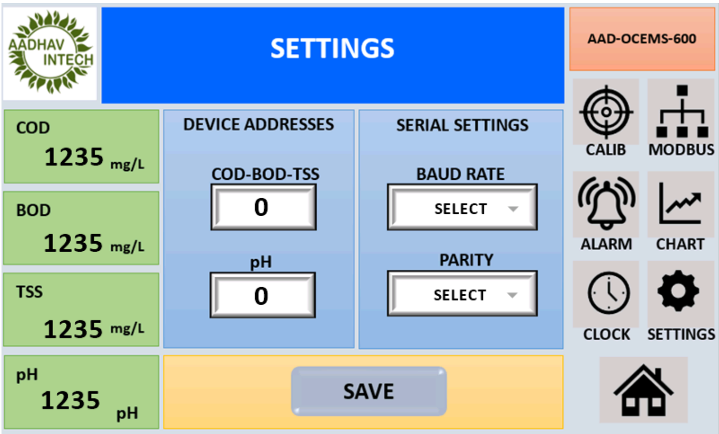


Figure 7 – Settings Screen

The Settings screen of the system is used to configure the communication parameters required for accurate sensor data reading user must enter the correct Modbus slave addresses for the COD-BOD-TSS and pH sensors to ensure proper identification and communication.

Additionally, the serial settings—such as baud rate and parity—must match the configuration set on the connected sensors. Once the device addresses and serial settings are correctly selected, pressing the "SAVE" button will store the configuration and initiate data communication.

Incorrect entries may result in failed or incorrect sensor readings. The settings screen also provides quick access to other system functions like calibration, Modbus, alarm setup, chart view, clock settings, and the main dashboard through clearly labelled icons on the right side.

Maintenance

Maintenance Guidelines:

- Regular maintenance of the AAD-OCEMS-600 system is essential to ensure accurate measurement, long-term reliability, and consistent performance. Although the system uses a reagent-free optical method that minimizes maintenance requirements, periodic checks and cleaning are recommended as part of preventive care.
- One of the most important maintenance tasks is cleaning the viper, which refers to the optical light path of the UV sensor. Over time, the light path may become contaminated with organic deposits, turbidity, or scale that can interfere with UV transmission and lead to inaccurate readings. To clean the viper, first power off the system, then carefully remove the sensor according to the manufacturer's instructions. Use a soft, lint-free cloth or swab with distilled water or a recommended sensor-safe cleaning solution to gently wipe the optical surfaces. Avoid abrasive materials or solvents that may scratch or damage the optical window.
- In addition to the viper cleaning, the electrode probes for pH and TSS should be checked weekly and cleaned as needed. The sample chamber or flow cell should be flushed periodically to prevent sediment or biofilm buildup. Sensor calibration should be performed at regular intervals or if a drift in readings is observed. Also, verify alarm settings and communication parameters monthly.
- The optical sensor and electrode probes should be inspected weekly for any signs of fouling, scaling, or dirt accumulation, especially in environments with high particulate or organic loads. Use a soft, lint-free cloth or appropriate cleaning solution (as recommended by the sensor manufacturer) to gently clean the sensor surfaces. Avoid using abrasive materials that could damage the optical components.

- The flow cell or sample chamber should also be flushed periodically to remove any sediment or biological growth. Calibration of sensors should be performed as per the calibration schedule or whenever drift in measurement is observed.
- Alarm thresholds and system settings should be reviewed monthly to ensure they align with operational and regulatory requirements.
- It is recommended to inspect electrical connections, power supply, and relay outputs regularly to avoid communication or control failures. In case of firmware updates or prolonged downtime, recheck all configuration parameters before restarting the system.
- Always turn off the power supply before performing any maintenance activities, and follow standard safety procedures. For detailed maintenance procedures, refer to the sensor-specific manuals provided by the manufacturer.